

Day 8 — Modern Lunar Science

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: phases as sun's illumination of moon; mention libration, tides.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pañcāṅga* (IAST: pañcāṅga; lit. “five-limbed Indian calendar (tithi, vāra, nakṣatra, yoga, karaṇa)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Phases re-explained.** Lunar phases = sunlight reflection from a 1737-km-radius sphere as the Earth-Moon-Sun angle changes. Synodic period of ~29.53 days. There is no “dark side” — only a *far side*, which receives the same sunlight, just not visible from Earth.
2. **Libration.** Earth-observers see ~59% of the Moon’s surface over time due to orbital eccentricity and axial tilt. Demo with rotating ball.
3. **Tidal lock.** Same face always visible to Earth — rotation period = orbital period = ~27.32 days (sidereal). This is the result of tidal forces over billions of years.
4. **Modern lunar exploration.** Mention briefly: Apollo missions (1969–72), Chandrayaan-1 (2008) discovered water-ice signatures, Chandrayaan-3 (2023) soft-landing near south pole, Artemis program (2024+).
5. **Bridge back to tradition.** Sūrya-siddhānta computes Moon’s distance and diameter approximately (Earth-Moon ~38000 yojanas ≈ 380000 km — close to modern 384400 km). The empirical precision is striking; the cosmological framework around it is what differs.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Modern Lunar Science.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).